Deuteron · 3-minute video script

355 spoken words · 9 visual scenes · finish speaking by 2:55, then hold for five seconds. Record one clean, continuous take with no background music. Read only the narration below; the timecodes and visual cues are for rehearsal.

0:00–0:12 | Slide 1: The compiler transformations we built

Deuteron compiles scheduled Emmy Kernel IR into PTX for the RTX 5090. Our contribution is guarded compiler passes that change memory reuse, parallel work, and tensor pipelines.

Cue: Open on the presenter, then the title. Point to reuse, parallel work, and tensor pipelines.

0:12–0:27 | Slide 2: Where our passes run

We validate IR and metadata; each rewrite checks index and ownership conditions. Early passes vectorize loops, forward reduction inputs, reorder recurrences, and fold split-K grids. Specialized lowering follows; LLVM supplies optimization and NVPTX code generation.

Cue: Follow the pipeline. Say the LLVM credit clearly.

0:27–0:44 | Slide 3: Reuse data before loading it again

Elementwise vectorization packs four values per lane. Reduction load forwarding keeps inputs in registers for the consumer instead of loading them again. Coalesced sum and square reductions use contiguous loads; long eligible loops add independent accumulators.

Cue: Trace the retained vector through reduction and consumer.

Deuteron · reductions and tensors

0:44–1:03 | Slide 4: Reshape parallel work

Cooperative dot reduction shares a repeated dot across lanes and combines partials with shuffles. Small-tree elimination removes redundant reduction stages. Split-K grid folding merges eligible K partitions into deeper work per block while preserving the original physical grid.

Cue: Show lane cooperation, then folded K work and inactive surplus blocks.

1:03–1:26 | Slide 5: Turn contractions into tensor work

Contraction tensorization recognizes a closed shared-memory multiply-add and emits compensated BF16x3 or TF32x3 products. Three high and residual products accumulate in FP32; omitting low times low makes this approximate. Shared-memory layout matches producers to MMA fragments. Separate FP16 storage uses explicit FP32 arithmetic.

Cue: Explain the three products. Keep the FP16 strip separate from tensor operands.

1:26–1:49 | Slide 6: Rewrite the pipeline feeding tensor cores

Packed register staging replaces a certified FP32 copy ring for BF16 lowering with vector prefetch, high and residual packing, and one used shared stage. Fragment pipelining loads the next MMA fragment early. Pipeline drain skips unused final prefetches. Other eligible paths retain asynchronous copies.

Cue: Compare the two paths. The packed BF16 path replaces the asynchronous ring.

Deuteron · attention and results

1:49–2:12 | Slide 7: Recognize and stage the attention recurrence

Staged online recurrence computes attention scores once per sharing group, then materializes unnormalized weights for mass and value accumulation. Blocked recurrence loads group adjacent keys. Shared-storage reuse overwrites dead operands only when warp ownership and dependency checks make it safe.

Cue: Follow scores to weights to mass and PV. Then show guarded storage reuse.

2:12–2:30 | Slide 8: Redistribute probability–value work

Warp-distributed vector PV reduction targets one warp and thirty-two columns. Four groups split the keys; each lane accumulates four adjacent columns. Shuffles combine the groups. The original output suffix is replayed, restoring the denominator before every column.

Cue: Trace four key shards into the first eight writing lanes.

2:30–2:55 | Slide 9: Measure the whole compiler on RTX 5090

On fifteen open workloads, geometric mean warm latency is three point seven five microseconds. Speedups are one point six six times over generated CUDA with NVCC, and one point one five times over the tested Triton baseline. All two hundred seventy calls passed the original checks. These are whole-compiler results with a limited Triton search.

Cue: Slow for the ratios. Hold the results slide silently from 2:55 to 3:00.

2:55–3:00: stop speaking; hold slide 9. Leave extra silence on the raw recording for editing. The full recording setup, pronunciation notes, optional cuts, and edit plan are in the companion Markdown files.